



Final Project

Fluid Dynamics SERA Competition Entry

Project Background

The SERA Fluid Dynamics Competition challenges students to design a 1.5 - 2.0 mL Eppendorf-style tube that can reliably contain liquids in microgravity. Students submit a 3D CAD model and a short design report explaining how their tube's geometry and microgravity adaptations (e.g., features that exploit surface tension and capillary action) keep fluids stable and easy to handle. Designs are judged for innovation, predicted functionality, manufacturability (with Form4 3D printer).

This final project can be submitted to the SERA competition when the competition portal opens in summer 2025. The top-scoring tubes will have the chance to be selected for flight on a Blue Origin New Shepard mission, where astronauts will film standardized maneuvers to observe how fluid behaves at three fill levels (0.5 mL, 1.0 mL, and max capacity). This in-flight testing will provide real microgravity performance data and a high-visibility showcase for the winning designs.

AI Policy

AI/LLMs should not be used to generate text for your final report. AI-detection algorithms may be run to validate the proposal was created in your own words.

Final Project Instructions

Competition Pre-Submission Package	2
Submission Checklist	2
Design Report Instructions	2
Grading Rubric	3
Final Presentation	4
Submission Checklist	4
Instructions	4
Grading Rubric	5

Competition Pre-Submission Package

30% of final grade

Submission Checklist

Deliverable	Submission Instructions
Design Report (≤ 5 pages, PDF)	Upload to Populi under the SERA Written Proposal assignment
CAD model file	Upload to Populi under the SERA Written Proposal assignment
Physical 3D-printed tube	Deliver to professor by the end of class (3:15pm) on Friday June 13, 2025

Design Report Instructions

Section

Guidance

Design Name

Give your tube a short design name

Fluid Dynamics in Microgravity

Explain how microgravity alters liquid behavior.

Explain why a conventional 1.5 mL tube must be re-imagined.

You may integrate diagrams (computer-generated or handdrawn) or equations to assist your explanation.

If external websites or documents are referenced, cite your sources.

Design Rationale for Microgravity

In an integrated narrative (no required subsections), describe all major features and explain how each mitigates the fluid-handling challenges outlined in Section 1. You may tie claims to sketches and renders, but focus on the *why*: how do your choices harness or counteract microgravity physics?

Design Evolution & Iterations

Chronicle your development process to reach a final tube design. For example, you can discuss discarded ideas, failed concepts, or how your thought process evolved throughout the term. Focus on the lessons you learned that drove your final design. Include sketches, tables, or a brief timeline as helpful.

Grading Rubric

Criterion	Excellent (A)	Good (B)	Satisfactory (C)	Poor (D/F)	Pts
Design Name	Memorable, relevant, and used consistently throughout report and files	Clear but less memorable or used inconsistently	Generic or confusing name	No name provided	2
Fluid Dynamics in Microgravity	Thorough, accurate explanation of altered forces; clearly justifies need to redesign tube; all sources cited	Accurate but lacks depth, minor citation or clarity issues	Basic description, limited justification	Inaccurate, superficial	8
Design Rationale for Microgravity	Integrated narrative links every major feature to specific fluid-physics challenge; strong “why” justification	Covers most features with partial linkage or evidence	Describes features but provides limited rationale	Minimal rationale or missing linkage to microgravity physics	15
Design Evolution & Iterations	Chronicles design process (ex: discarded ideas); insightful reflection on failures and lessons	Describes iterations with some reflection	Mentions iterations briefly, little analysis	No development history or lessons	10
Technical Writing & Organization	Professional tone; logical flow within three required sections	Generally clear, minor edits needed	Adequate but wordy or error-prone	Disorganized or unclear	5
CAD File	CAD File matches the description of changes in the design file. All novel features from the design file are present.	Minor differences between design description and file	CAD file minimally reflects design described in report.	Unusable or missing CAD	30
3D Print Quality	Matches CAD file. Design is capable of holding fluid, such as water without leakage.	Small dimensional or finish flaws; minor fluid leakage.	Noticeable warping/print failure; water is not reasonably contained	Off-spec, substantial water leakage, or 3D print is missing	30

= 100 pts total

Final Presentation

20% of final grade

Submission Checklist

Deliverable	Submission Instructions
Slide Deck	Upload to Populi under the SERA Proposal Presentation assignment
Live Presentation	Deliver during class on Friday June 13, 2025

Instructions

Deliver a tightly focused 5- to 6-minute talk that walks the class, and the filming crew, through your tube's CAD visuals and design rationale. Use the instructor-provided slide template to weave a clear narrative that highlights the challenges of manipulating fluids in microgravity and argues for why your tube design should replace the standard 1.5mL microfuge tube.

When	Friday, June 13, 2:00 – 3:15 pm (in-class, filmed by UATX Comms)
Where	UATX Library
Length	5-6 minutes + 2 min Q&A from classmates
Slides	Use the instructor-provided slide template

Grading Rubric

Criterion	Excellent (A)	Good (B)	Satisfactory (C)	Poor (D/F)	Pts
Description of Motivating Problem: Fluid Behavior in Microgravity	Clearly and concisely explains how the absence of gravity reshapes surface tension, capillary action, and fluid containment challenges; connects these physics principles to the need for a redesigned tube	Covers the main microgravity effects with minor gaps or vague links to design need	States basic facts about microgravity but with limited depth or unclear relevance	Little-to-no explanation or inaccuracies about fluid behavior in space	25
Description of Innovation	Justification for how the new geometry/feature(s), improve on a standard 1.5 mL microfuge tube for microgravity	Highlights several improvements with partial evidence or connection to microgravity constraints	Identifies at least one innovation but provides limited rationale or supporting details	Minimal or no innovation presented; improvements unsubstantiated	35
Technical Clarity & Structure	Logical flow, crisp explanations, smooth transitions; stays within 5–6 min	Generally clear, minor digressions or slight time variance	Some disorganization or jargon; pacing issues	Disjointed, confusing, or off-time	15
Visual Quality & Template Use	Template followed perfectly; clean, readable slides with high-quality visuals and annotations	Minor template or readability issues	Crowded or stylistically inconsistent slides	Difficult to read; ignores template	10
Delivery & Time Management	Confident, engaging, well-paced; finishes within allotted time	Mostly confident, minor pacing or reading	Reads heavily from notes; slight over/under time	Monotone, unclear, or major time violations	10
Q&A / Audience Engagement	Responds precisely to questions; actively engages peers	Answers questions mostly accurately	Vague or partial answers	Inaccurate answers or unable to respond	5

= 100 pts total